

iapws

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NAME

iapws – library for light and heavy water properties according to IAPWS.

LIBRARY

iapws library (*libiapws*, *-liapws*)

SYNOPSIS

Fortran **use iapws**

C **#include "iapws.h"**

Python **import pyiapws**

DESCRIPTION

Numerical implementation for reports:

- R2-83
 - [x] Tc in H2O and D2O
 - [x] pc in H2O and D2O
 - [x] rhoc in H2O and D2O
- G7-04
 - [x] kH
 - [x] kD
- R7-97
 - [x] Region 1
 - [] Region 2
 - [] Region 3
 - [x] Region 4
 - [] Region 5
- R11-24:
 - [x] Kw

NOTES

dp stands for double precision and it is an alias to real64 from the iso_fortran_env module.

Water phases:

- | | |
|---|----------------|
| l | liquid |
| v | vapor |
| c | super critical |
| s | saturation |
| n | unknown |

References

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- [11] H. Arcis et al., "Revised Parameters for the IAPWS Formulation for the Ionization Constant of Water Over a Wide Range of Temperatures and Densities, Including Near-Critical Conditions," Journal of Physical and Chemical Reference Data, vol. 53, no. 2, p. 23103, May 2024, doi: 10.1063/5.0198792.

EXAMPLES

Example in Fortran:

```

program example_in_f
  use iapws__common, only: dp, int32
  use iapws
  implicit none(type,external)

  integer(int32) :: i, ngas
  real(dp) :: T(1), kh_res(1), kd_res(1), wp_res(1), p(1)
  real(dp) :: Ts(7), ps(7)
  real(dp) :: x(3), y(3)
  integer(int32) :: r(3)
  character(len=1) :: s(3)
  character(len=2) :: gas = "O2"
  integer(int32) :: heavywater = 0
  type(gas_type), pointer :: gases_list(:)
  character(len=:), pointer :: gases_str

  print *, '##### IAPWS VERSION #####'
  print *, "version ", version()

  print *, '##### IAPWS R2-83 #####'
```

```

print "(a, f10.3, a)", "Tc in h2o=", Tc_H2O, " k"
print "(a, f10.3, a)", "pc in h2o=", pc_H2O, " mpa"
print "(a, f10.3, a)", "rhoc in h2o=", rhoc_H2O, " kg/m3"

print "(a, f10.3, a)", "Tc in D2O=", Tc_D2O, " k"
print "(a, f10.3, a)", "pc in D2O=", pc_D2O, " mpa"
print "(a, f10.3, a)", "rhoc in D2O=", rhoc_D2O, " kg/m3"
print *, ''

print *, '##### IAPWS G7-04 #####'
! Compute kh and kd in H2O
T(1) = 25.0_dp + 273.15_dp
call kh(T, gas, heavywater, kh_res)
print "(A10, 1X, A10, 1X, A2, F10.1, A, 4X, A3, SP, F10.4)", "Gas=", gas, "T="

call kd(T, gas, heavywater, kd_res)
print "(A10, 1X, A10, 1X, A2, F10.1, A, 4X, A3, SP, F15.4)", "Gas=", gas, "T="

! Get and print available gases
heavywater = 0
ngas = ngases(heavywater)
gases_list => null()
gases_list => gases(heavywater)
gases_str => gases2(heavywater)
print *, "Gases in H2O: ", ngas
print *, gases_str
do i=1, ngas
print *, gases_list(i)%gas
enddo

heavywater = 1
ngas = ngases(heavywater)
gases_list => null()
gases_list => gases(heavywater)
gases_str => gases2(heavywater)
print *, "Gases in D2O: ", ngas
print *, gases_str
do i=1, ngas
print *, gases_list(i)%gas
enddo

print *, '##### IAPWS R7-97 #####'
! Compute ps from Ts.
Ts(:) = [-1.0_dp, 25.0_dp, 100.0_dp, 200.0_dp, 300.0_dp, 360.0_dp, 374.0_dp]
Ts(:) = Ts(:) + 273.15_dp
call psat(Ts, ps)

do i=1, size(Ts)
print "(SP, F23.3, A3, 4X, F23.3, A3)", Ts(i), "K", ps(i), "MPa"
end do

! Compute Ts from ps
call Tsat(ps, Ts)
do i=1, size(Ts)

```

```

print "(SP, F23.3, A3, 4X, F23.3, A3)", Ts(i), "K", ps(i), "MPa"
end do

! Compute water properties at 280°C/8 Mpa
p(1) = 8.0_dp
T(1) = 273.15_dp + 280.0_dp
call wp(p, T, "v", wp_res)
print "(A5, F23.16, X, A)", "v(8MPa,280°C)=", wp_res(1)*1000.0_dp, "L/kg"

! Compute region and phase
x = [8.0_dp, 4.0_dp, 6.0_dp ]
y = [553.15_dp, 1200.0_dp, 2000.0_dp]
call wr(x, y, r)
call wph(x, y, s)
print *, r
print *, s

end program example_in_f

```

Example in C

```

#include <string.h>
#include <stdio.h>
#include "iapws.h"

int main(void){
double T = 25.0 + 273.15; /* in C*/
double p; /* p in Mpa */
char *gas = "O2";
double kh, kd, wp_res;
char **gases_list;
char *gases_str;
int ngas;
int i;
int heavywater = 0;
double x[3]= {8.0, 4.0, 6.0 };
double y[3] = {553.15, 1200.0, 2000.0};
int r[3];
char s[3];

printf("%s0, "##### IAPWS VERSION #####
printf("version %s0, iapws_version());

printf("%s0, "##### IAPWS R2-83 #####
printf("%s %10.3f %s0, "Tc in H2O", iapws_r283_Tc_H2O, "K");
printf("%s %10.3f %s0, "pc in H2O", iapws_r283_pc_H2O, "MPa");
printf("%s %10.3f %s0, "rhoc in H2O", iapws_r283_rhoc_H2O, "kg/m3");

printf("%s %10.3f %s0, "Tc in D2O", iapws_r283_Tc_D2O, "K");
printf("%s %10.3f %s0, "pc in D2O", iapws_r283_pc_D2O, "MPa");
printf("%s %10.3f %s0, "rhoc in D2O", iapws_r283_rhoc_D2O, "kg/m3");

printf("0);

```

```

printf("%s0, "##### IAPWS G7-04 #####
/* Compute kh and kd in H2O*/
iapws_g704_kh(&T, gas, heavywater, &kh, strlen(gas), 1);
printf("Gas=%sT=%fKkh=%+10.4f0, gas, T, kh);

iapws_g704_kd(&T, gas, heavywater, &kd, strlen(gas), 1);
printf("Gas=%sT=%fKkd=%+15.4f0, gas, T, kd);

/* Get and print the available gases */
ngas = iapws_g704_ngases(heavywater);
gases_list = iapws_g704_gases(heavywater);
gases_str = iapws_g704_gases2(heavywater);
printf("Gases in H2O: %d0, ngas);
printf("%s0, gases_str);
for(i=0; i<ngas; i++){
    printf("%s0, gases_list[i]);
}

heavywater = 1;
ngas = iapws_g704_ngases(heavywater);
gases_list = iapws_g704_gases(heavywater);
gases_str = iapws_g704_gases2(heavywater);
printf("Gases in D2O: %d0, ngas);
printf("%s0, gases_str);
for(i=0; i<ngas; i++){
    printf("%s0, gases_list[i]);
}

printf("%s0, "##### IAPWS R7-97 #####
double Ts[7] = {-1.0, 25.0, 100.0, 200.0, 300.0, 360.0, 374.0};
double ps[7] = {1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0};
for(i=0; i<7; i++){
    Ts[i] = Ts[i] + 273.15;
}
iapws_r797_psat(7, Ts, ps);

for(i=0; i<7; i++){
    printf("%+23.3f %s %+23.3f %s0, Ts[i], "K", ps[i], "MPa");
}

iapws_r797_Tsat(7, ps, Ts);
for(i=0; i<7; i++){
    printf("%+23.3f %s %+23.3f %s0, Ts[i], "K", ps[i], "MPa");
}

T = 273.15 + 280.0;
p = 8.0;
iapws_r797_wp(&p, &T, "v", &wp_res, 1, 1);
printf("v(8MPa,280Â°C) = %+23.16f L/kg0, wp_res * 1000.0);

iapws_r797_wr(x, y, r, 3);
iapws_r797_wph(x, y, s, 3);
for(i=0; i<3; i++){
    printf("%i", r[i]);
}

```

```

}
printf("0);
for(i=0; i<3; i++){
    printf("%c", s[i]);
}
printf("0);

return 0;
}

```

Example in Python

```

r"""Example in python"""
import sys
sys.path.insert(0, "../py/src/")
import array
import numpy as np
import matplotlib.pyplot as plt
import pyiapws

print("##### IAPWS VERSION #####")
print(pyiapws.__version__)

print("##### IAPWS R2-83 #####")
print("Tc in H2O", pyiapws.Tc_H2O, "K")
print("pc in H2O", pyiapws.pc_H2O, "MPa")
print("rhoc in H2O", pyiapws.rhoc_H2O, "kg/m3")

print("Tc in D2O", pyiapws.Tc_D2O, "K")
print("pc in D2O", pyiapws.pc_D2O, "MPa")
print("rhoc in D2O", pyiapws.rhoc_D2O, "kg/m3")

print("")

print("##### IAPWS G7-04 #####")
gas = "O2"
T = array.array("d", (25.0+273.15,))

# Compute kh and kd in H2O
heavywater = False
k = pyiapws.kh(T, "O2", heavywater)
print(f"Gas={gas}T={T[0]}Kkh={k[0]:+10.4f}0)

k = pyiapws.kd(T, "O2", heavywater)
print(f"Gas={gas}T={T[0]}Kkd={k[0]:+10.4f}0)

# Get and print the available gases
heavywater = False
gases_list = pyiapws.gases(heavywater)
gases_str = pyiapws.gases2(heavywater)
ngas = pyiapws.ngases(heavywater)
print(f"Gases in H2O: {ngas}")
print(gases_str)
for gas in gases_list:
    print(gas)

```

```

heavywater = True
gases_list = pyiapws.gases(heavywater)
gases_str = pyiapws.gases2(heavywater)
ngas = pyiapws.ngases(heavywater)
print(f"Gases in D2O: {ngas}")
print(gases_str)
for gas in gases_list:
    print(gas)

style = {"marker": ".", "ls": "", "ms": 2}
T_KELVIN = 273.15
T = np.linspace(0.0, 360.0, 1000) + 273.15

solvent = {True: "D2O", False: "H2O"}

print("Generating plot for kh")
kname = "kh"
for HEAVYWATER in (False, True):
    print(solvent[HEAVYWATER])
    fig = plt.figure()
    ax = fig.add_subplot()
    ax.grid(visible=True, ls=':')
    ax.set_xlabel("T / Å°K")
    ax.set_ylabel("ln (kh/1GPa)")
    gases = pyiapws.gases(HEAVYWATER)
    for gas in gases:
        k = pyiapws.kh(T, gas, HEAVYWATER) / 1000.0
        ln_k = np.log(k)
        ax.plot(T, ln_k, label=gas, **style)
    ax.legend(ncol=3)
    fig.savefig(f"..media/g704-{kname:s}_{solvent[HEAVYWATER]}.png", dpi=100)

print("Generating plot for kd")
kname = "kd"
for HEAVYWATER in (False, True):
    print(solvent[HEAVYWATER])
    fig = plt.figure()
    ax = fig.add_subplot()
    ax.grid(visible=True, ls=':')
    ax.set_xlabel("T / Å°K")
    ax.set_ylabel("ln kd")
    gases = pyiapws.gases(HEAVYWATER)
    for gas in gases:
        k = pyiapws.kd(T, gas, HEAVYWATER)
        ln_k = np.log(k)
        ax.plot(T, ln_k, label=gas, **style)
    ax.legend(ncol=3)
    fig.savefig(f"..media/g704-{kname:s}_{solvent[HEAVYWATER]}.png", dpi=100)

print("##### IAPWS R7-97 #####")
Ts = np.asarray([-1.0, 25.0, 100.0, 200.0, 300.0, 360.0, 374.0])
Ts = Ts + 273.15

```



```

ps = pyiapws.psat(Ts)
for i in range(Ts.size):
    print(f"{Ts[i]:23.3f} K {ps[i]:23.3f} MPa.")

Ts = pyiapws.Tsat(ps)
for i in range(Ts.size):
    print(f"{Ts[i]:23.3f} K {ps[i]:23.3f} MPa.")

fig = plt.figure()
ax = fig.add_subplot()
ax.grid(visible=True, ls=':')
ax.set_xlabel("Ts /K")
ax.set_ylabel("ps /MPa")
Ts = np.linspace(0.0, 370.0, 500)
Ts = Ts + 273.15

ps = pyiapws.psat(Ts)
ax.plot(Ts, ps, "r-", label="ps(Ts)")

Ts = pyiapws.Tsat(ps)
ax.plot(Ts, ps, "b--", label="Ts(ps)")

ax.legend()
fig.savefig(f"../media/r797-r4.png", dpi=100, format="png")

T = 280.0 + 273.15
p = 8.0
res = pyiapws.wp(p, T, "v")*1000.0
print(f"v(8MPa,280Â°C) = {res:+23.16f} L/kg)")

x = np.asarray([8.0, 4.0, 6.0 ])
y = np.asarray([553.15, 1200.0, 2000.0])
r=pyiapws.wr(x, y)
s=pyiapws.wph(x, y)
print(r)
print(s)

plt.show()

```

SEE ALSO

iapws(3) iapws_cli(1) iapws_kh(3) ciaaw(3), codata(3),

NAME

iapws – Compute light and heavy water properties.

SYNOPSIS

iapws *SUBCOMMAND* [*OPTION*]...

DESCRIPTION

iapws is a command line interface for computing properties of light and heavy water according to IAPWS.

SUBCOMMANDS

Valid subcommands are:

- + kh** Compute the Henry's constant for gases in H₂O or D₂O. The default behavior is to compute the constant kH for O₂ at 25°C. See options.
- + kd** Compute the vapor-liquid distribution constant for gases in H₂O or D₂O. The default behavior is to compute the constant kD for H₂ at 25°C. See options.
- + psat** Compute the saturation pressure. The default behavior is to compute psat at 25°C. See options.
- + Tsat** Compute the saturation temperature. The default behavior is to compute Tsat at 1 bar. See options.
- + wp** Compute water properties for regions 1 to 5. The default behavior is to compute the properties at 25°C and 1 bar. **WARNING:** Currently, only region 1 is supported.

Their syntax is:

- + kh** [*OPTION*]...
- + kd** [*OPTION*]...
- + psat** [*OPTION*]...
- + Tsat** [*OPTION*]...
- + wp** [*OPTION*]...

OPTIONS

kh:

- temperature, -T TEMPERATURE...**
Temperature in °C. Default to 25°C.
- fugacity, -f FUGACITY...**
Liquid-phase fugacity in MPa. Default to 1 bar.
- gases, -g GAS...**
Gases for which to compute kH. Default to O₂.
- D2O** Set heavywater as the solvent.
- listgases**
Display available gases for computing kH.

kd:

- temperature, -T TEMPERATURE...**
Temperature in °C. Default to 25°C.

--x2, -x x2...
Molar fraction of gas in water. Default to 1e-9.

--gases, -g GAS...
Gases for which to compute kD. Default to H2.

--D2O, Set heavywater as the solvent.

--listgases
Display available gases for computing kD.

psat:

--temperature, -T TEMPERATURE...
Temperature in $^{\circ}\text{C}$. Default to 25 $^{\circ}\text{C}$.

Tsat:

--pressure, -p PRESSURE...
Pressure in bar. Default to 1 bar.

wp:

--temperature, -T TEMPERATURE...
Temperature in $^{\circ}\text{C}$. Default to 25 $^{\circ}\text{C}$.

--pressure, -p PRESSURE...
Pressure in bar. Default to 1 bar.

all:

--usage, -u
Show usage text and exit.

--help, -h
Show help text and exit.

--verbose, -V
Display additional information.

--version, -v
Show version information and exit.

NOTES

You may replace the default options from a file if your first options begin with @file. Initial options will then be read from the "response file" "file.rsp" in the current directory.

If "file" does not exist or cannot be read, then an error occurs and the program stops. Each line of the file is prefixed with "options" and interpreted as a separate argument. The file itself may not contain @file arguments. That is, it is not processed recursively.

For more information on response files see https://urbanjost.github.io/M_CLI2/set_args.3m_cli2.html

EXAMPLE

Minimal example

```
iapws kh -T 25,100 -f 1,0.2 -g O2,H2
iapws kd -T 25,100 -x2 1d-9,1d-6 -g O2,H2
iapws wp -T 25,100 -p 0.035,1.0
```

SEE ALSO

iapws(3) iapws_cli(1) iapws_kh(3) ciaaw(3), codata(3),

NAME

kh – function

LIBRARYiapws library (*libiapws*, *-liapws*)**SYNOPSIS**

```

Fortran      pure subroutine kh(T,gas,heavywater,k)
               real(dp), intent(in), contiguous :: T(:)
               character(len=*), intent(in) :: gas
               integer(int32), intent(in) :: heavywater
               real(dp), intent(out), contiguous :: k(:)

C            void iapws_g704_kh(double *T,char *gas,int heavywater,double *k,int size_gas,size_t
               size_T)

Python       kh(T:np.ndarray,gas:str,heavywater:bool=False)->Union[np.ndarray,float]:

```

DESCRIPTION

Computes the Henry constant kH in MPa for a given temperature T .

Arguments:

<i>T</i> (:)	Temperature in K.
<i>gas</i>	Gas.
<i>heavywater</i>	Flag if D2O (1) is used or H2O (0).
<i>k</i> (:)	Henry constant in MPa.

RETURN VALUE

k is filled with NaN if gas is not found.

EXAMPLES

Fortran:

```

use iapws
real(dp) :: T(1), kH(1)
T(1) = 25.0_dp + 273.15_dp
call kh(T, "O2", 0, kH)

```

C:

```

#include "iapws.h"
double T = 25.0 + 273.15;
double kH;
iapws_g704_kh(&T, "O2", 0, &kH, strlen(gas), 1);

```

Python:

```

import numpy as np
import pyiapws
T = np.asarray([25.0])

```

```
    kH = pyiapws.kh(T+273.15, "O2", False)
    print(kH)
```

SEE ALSO

iapws(3) iapws_cli(1) iapws_kh(3) ciaaw(3), codata(3),

NAME

kh – function

LIBRARYiapws library (*libiapws*, *-liapws*)**SYNOPSIS**

```

Fortran      pure subroutine kd(T,gas,heavywater,k)
              real(dp), intent(in), contiguous :: T(:)
              character(len=*), intent(in) :: gas
              integer(int32), intent(in) :: heavywater
              real(dp), intent(out), contiguous :: k(:)

C            void iapws_g704_kd(double *T,char *gas,int  heavywater,double *k,int  size_gas,size_t
              size_T)

Python      kd(T:np.ndarray,gas:str,heavywater:bool=False)->Union[np.ndarray,float]:

```

DESCRIPTION

Computes the Henry constant kH in MPa for a given temperature T . Compute the vapor-liquid constant kD for a given temperature T .

Arguments:

$T(:)$	Temperature in K.
<i>gas</i>	Gas.
<i>heavywater</i>	Flag if D2O (1) is used or H2O (0).
$k(:)$	Vapor-liquid constant in MPa.

RETURN VALUE

k is filled with NaN if gas is not found.

EXAMPLES

Fortran:

```

use iapws
real(dp) :: T(1), kD(1)
T(1) = 25.0_dp + 273.15_dp
call kd(T, "O2", 0, kD)

```

C:

```

#include "iapws.h"
double T = 25.0 + 273.15;
double kD;
iapws_g704_kd(&T, "O2", 0, &kD, strlen(gas), 1);

```

Python:

```

import numpy as np
import pyiapws

```

```
T = np.asarray([25.0])
kD = pyiapws.kd(T+273.15, "O2", False)
print(kD)
```

SEE ALSO

iapws(3) iapws_cli(1) iapws_kh(3) ciaaw(3), codata(3),